

ENV 790.30 - Time Series Analysis for Energy Data | Spring 2026

Assignment 2 - Due date 01/27/26

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Submission Instructions

You should open the .rmd file corresponding to this assignment on RStudio. The file is available on our class repository on Github.

Once you have the file open on your local machine the first thing you will do is rename the file such that it includes your first and last name (e.g., “LuanaLima_TSA_A02_Sp26.Rmd”). Then change “Student Name” on line 4 with your name.

Then you will start working through the assignment by **creating code and output** that answer each question. Be sure to use this assignment document. Your report should contain the answer to each question and any plots/tables you obtained (when applicable).

When you have completed the assignment, **Knit** the text and code into a single PDF file. Submit this pdf using Sakai.

R packages

R packages needed for this assignment: “forecast”, “tseries”, and “dplyr”. Install these packages, if you haven’t done yet. Do not forget to load them before running your script, since they are NOT default packages.\

```
#Load required package
library(forecast)
library(ggplot2)
library(dplyr)
library(cowplot)
library(tseries)
library(readxl)
```

Data set information

Consider the data provided in the spreadsheet “Table_10.1_Renewable_Energy_Production_and_Consumption_by_Source” on our **Data** folder. The data comes from the US Energy Information and Administration and corresponds to the December 2025 Monthly Energy Review. The spreadsheet is ready to be used. Refer to the file “M2_ImportingData_XLSX.Rmd” in our Lessons folder for instructions on how to read .xlsx files.

```
#Importing data set
energy_data <- read_excel(
  path = "~/Documents/ENVIRON 797 /Exercise_and_Assignments/Class_Exercise_and_Assignments/Data/Table_10.1_Renewable_Energy_Production_and_Consumption_by_Source.xlsx",
  skip = 12,
  sheet = "Monthly Data",
  col_names = FALSE
)
```

```
## New names:
## * ' -> '...1'
## * ' -> '...2'
## * ' -> '...3'
## * ' -> '...4'
## * ' -> '...5'
## * ' -> '...6'
## * ' -> '...7'
## * ' -> '...8'
## * ' -> '...9'
## * ' -> '...10'
## * ' -> '...11'
## * ' -> '...12'
## * ' -> '...13'
## * ' -> '...14'
```

```
#Extract column names from row 11
```

```
read_col_names <- read_excel(
  path = "~/Documents/ENVIRON 797 /Exercise_and_Assignments/Class_Exercise_and_Assignments/Data/Table_1",
  skip = 10,
  n_max = 1,
  sheet = "Monthly Data",
  col_names = FALSE
)
```

```
## New names:
## * ' -> '...1'
## * ' -> '...2'
## * ' -> '...3'
## * ' -> '...4'
## * ' -> '...5'
## * ' -> '...6'
## * ' -> '...7'
## * ' -> '...8'
## * ' -> '...9'
## * ' -> '...10'
## * ' -> '...11'
## * ' -> '...12'
## * ' -> '...13'
## * ' -> '...14'
```

```
#Assign column names
```

```
colnames(energy_data) <- as.character(unlist(read_col_names))
```

```
head(energy_data)
```

```
## # A tibble: 6 x 14
##   Month      'Wood Energy Production' 'Biofuels Production'
##   <dtm>                                <dbl> <chr>
## 1 1973-01-01 00:00:00                130. Not Available
## 2 1973-02-01 00:00:00                117. Not Available
## 3 1973-03-01 00:00:00                130. Not Available
## 4 1973-04-01 00:00:00                125. Not Available
```

```
## 5 1973-05-01 00:00:00          130. Not Available
## 6 1973-06-01 00:00:00          125. Not Available
## # i 11 more variables: 'Total Biomass Energy Production' <dbl>,
## #   'Total Renewable Energy Production' <dbl>,
## #   'Hydroelectric Power Consumption' <dbl>,
## #   'Geothermal Energy Consumption' <dbl>, 'Solar Energy Consumption' <chr>,
## #   'Wind Energy Consumption' <chr>, 'Wood Energy Consumption' <dbl>,
## #   'Waste Energy Consumption' <dbl>, 'Biofuels Consumption' <chr>,
## #   'Total Biomass Energy Consumption' <dbl>, ...
```

Question 1

You will work only with the following columns: Total Biomass Energy Production, Total Renewable Energy Production, Hydroelectric Power Consumption. Create a data frame structure with these three time series only. Use the command `head()` to verify your data.

```
#Select the columns for this analysis
energy_subset <- energy_data %>%
  select(`Total Biomass Energy Production`,
         `Total Renewable Energy Production`,
         `Hydroelectric Power Consumption`)

#Convert data to numeric
energy_subset <- energy_subset %>%
  mutate(across(everything(), as.numeric))

head(energy_subset)
```

```
## # A tibble: 6 x 3
##   Total Biomass Energy Production~1 Total Renewable Ener~2 Hydroelectric Power ~3
##           <dbl>           <dbl>           <dbl>
## 1           130.           220.           89.6
## 2           117.           197.           79.5
## 3           130.           219.           88.3
## 4           126.           209.           83.2
## 5           130.           216.           85.6
## 6           126.           208.           82.1
## # i abbreviated names: 1: 'Total Biomass Energy Production',
## #   2: 'Total Renewable Energy Production',
## #   3: 'Hydroelectric Power Consumption'
```

Question 2

Transform your data frame in a time series object and specify the starting point and frequency of the time series using the function `ts()`.

```
#The data starts on January 1973
energy_ts <- ts(energy_subset, start = c(1973, 1), frequency = 12)
head(energy_ts)
```

```
##           Total Biomass Energy Production Total Renewable Energy Production
```

```
## Jan 1973          129.787          219.839
## Feb 1973          117.338          197.330
## Mar 1973          129.938          218.686
## Apr 1973          125.636          209.330
## May 1973          129.834          215.982
## Jun 1973          125.611          208.249
##      Hydroelectric Power Consumption
## Jan 1973          89.562
## Feb 1973          79.544
## Mar 1973          88.284
## Apr 1973          83.152
## May 1973          85.643
## Jun 1973          82.060
```

Question 3

Compute mean and standard deviation for these three series.

```
#Compute mean and standard deviation for each of the three series
energy_mean <- apply(energy_ts, 2, mean, na.rm = TRUE)
energy_sd   <- apply(energy_ts, 2, sd,   na.rm = TRUE)
energy_stats <- data.frame(Mean = energy_mean, SD = energy_sd)
energy_stats
```

```
##              Mean      SD
## Total Biomass Energy Production  286.04893  96.21209
## Total Renewable Energy Production 409.19521 151.42232
## Hydroelectric Power Consumption   79.35682  14.12020
```

Question 4

Display and interpret the time series plot for each of these variables. Try to make your plot as informative as possible by writing titles, labels, etc. For each plot add a horizontal line at the mean of each series in a different color.

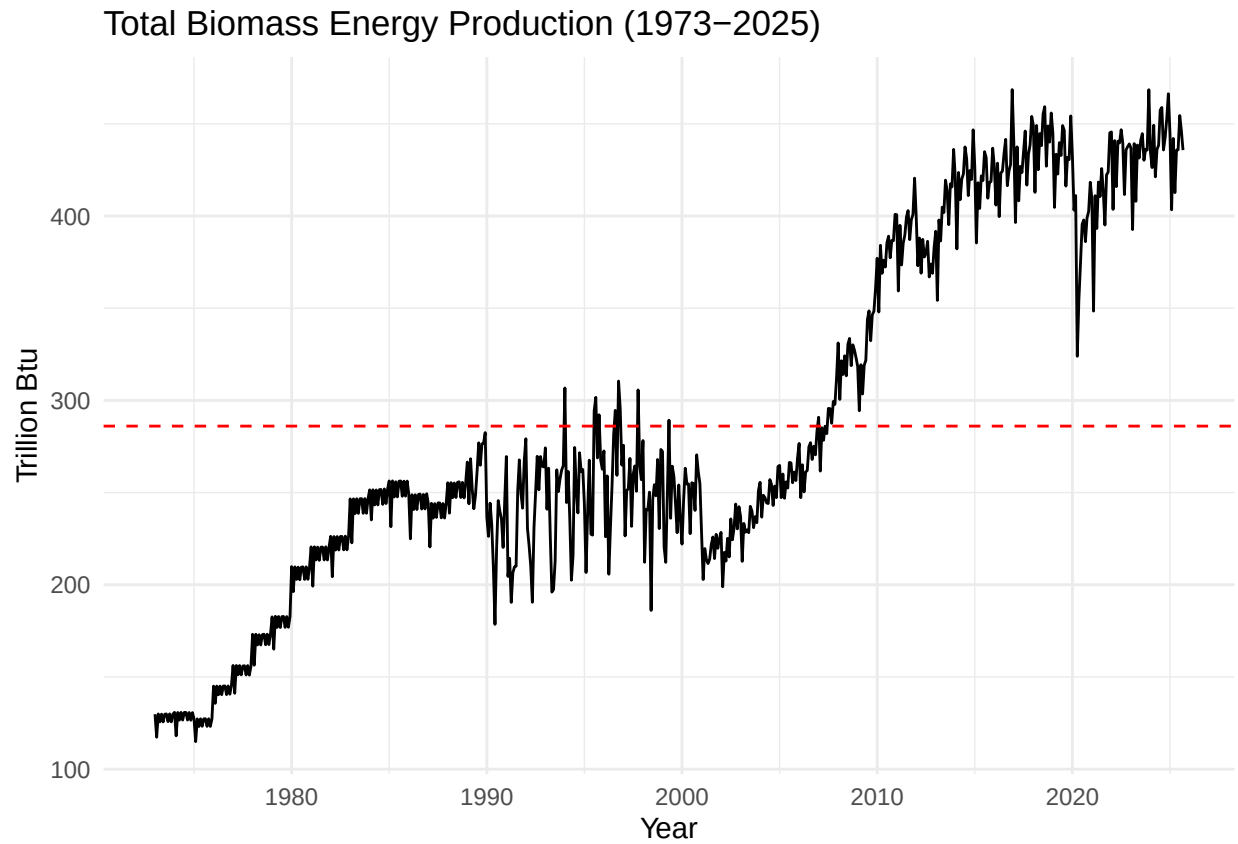
```
#Total Biomass Energy Production Plot
plot_biomass <- autoplot(energy_ts[, "Total Biomass Energy Production"]) +
  geom_hline(yintercept = energy_mean[1], color = "red", linetype = "dashed") +
  labs(title = "Total Biomass Energy Production (1973-2025)",
       x = "Year", y = "Trillion Btu") +
  theme_minimal()

#Total Renewable Energy Plot
plot_renewable <- autoplot(energy_ts[, "Total Renewable Energy Production"]) +
  geom_hline(yintercept = energy_mean[2], color = "blue", linetype = "dashed") +
  labs(title = "Total Renewable Energy Production (1973-2025)",
       x = "Year", y = "Trillion Btu") +
  theme_minimal()

#Hydroelectric Power Consumption Plot
plot_hydro <- autoplot(energy_ts[, "Hydroelectric Power Consumption"]) +
  geom_hline(yintercept = energy_mean[3], color = "darkorange", linetype = "dashed") +
```

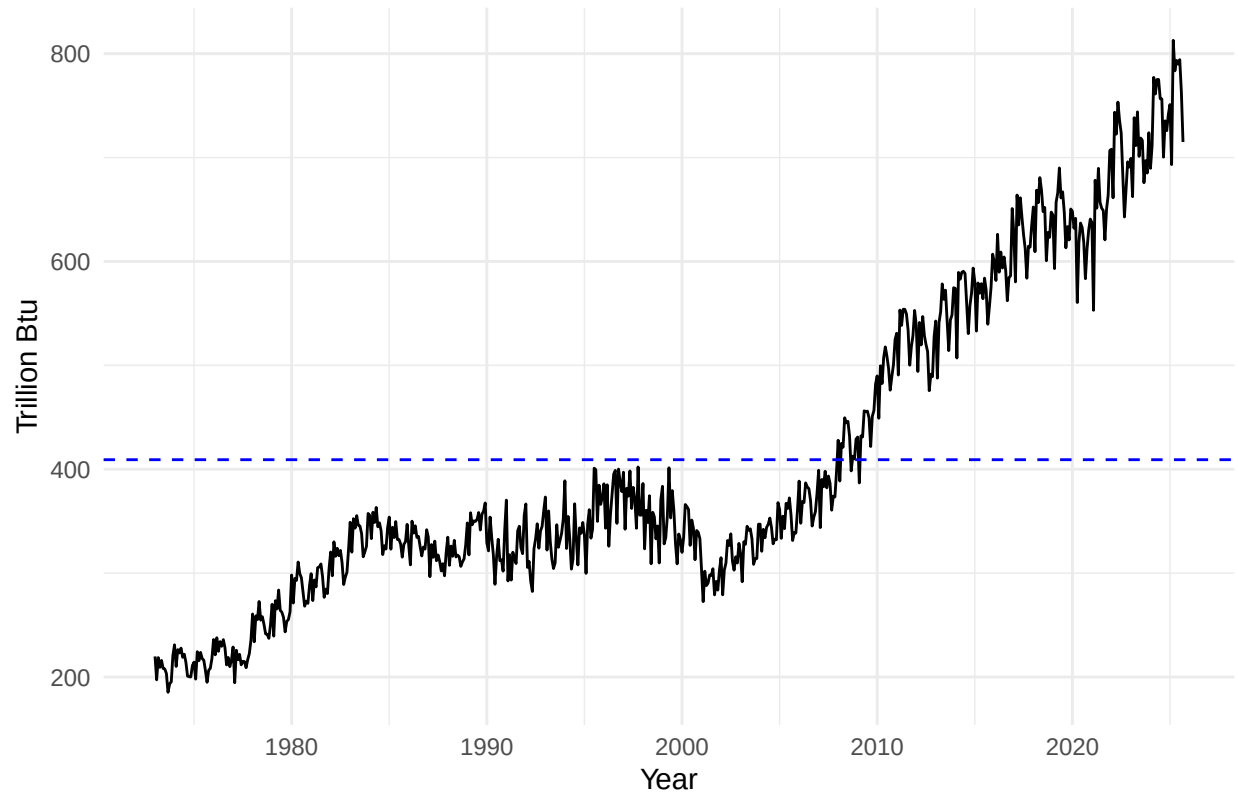
```
labs(title = "Hydroelectric Power Consumption (1973-2025)",  
      x = "Year", y = "Trillion Btu") +  
theme_minimal()
```

plot_biomass



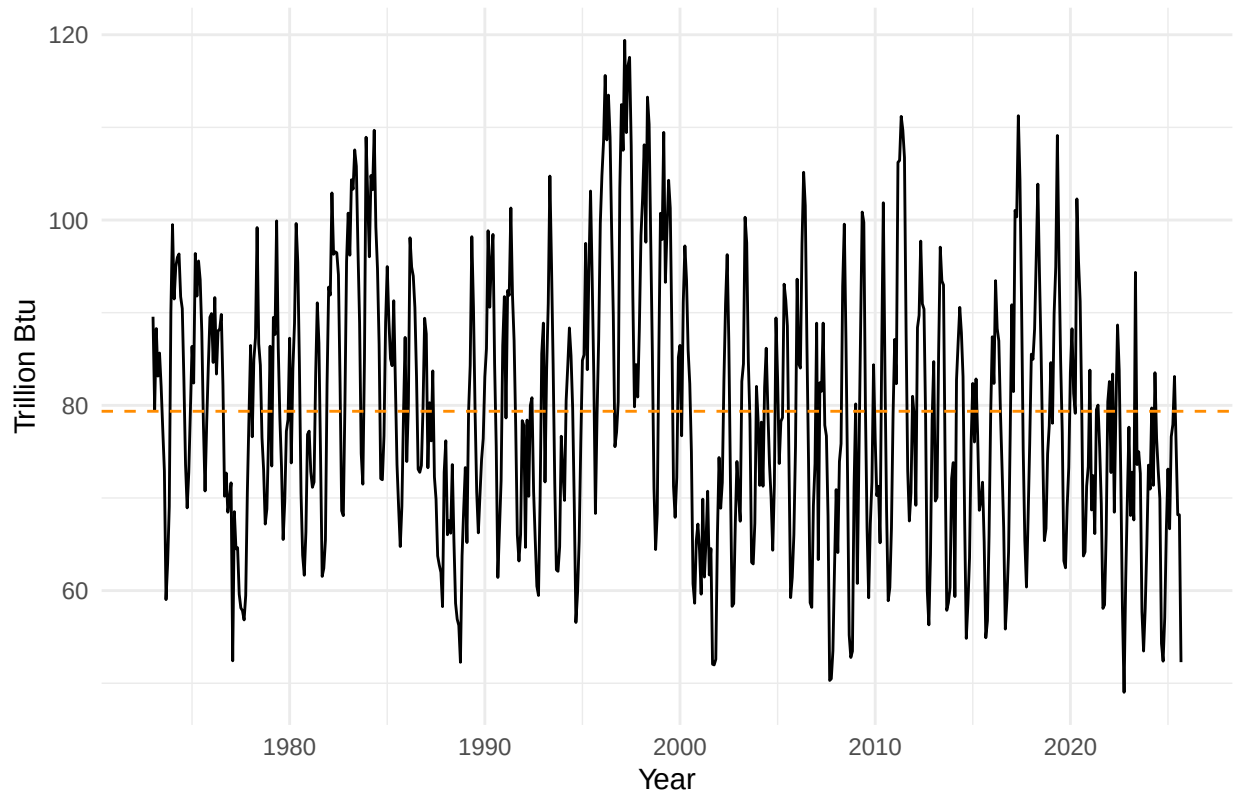
plot_renewable

Total Renewable Energy Production (1973–2025)



plot_hydro

Hydroelectric Power Consumption (1973–2025)



Answer: The biomass and renewable energy plots show a clear increasing trend, with most recent values sitting well above the mean. Hydroelectric power is different, it fluctuates around the mean with no clear long-term trend, but shows a repeating seasonal pattern.

Question 5

Compute the correlation between these three series. Are they significantly correlated? Explain your answer.

```
#Pearson correlation among the three series
```

```
energy_corr <- cor(energy_subset, use = "complete.obs")
energy_corr
```

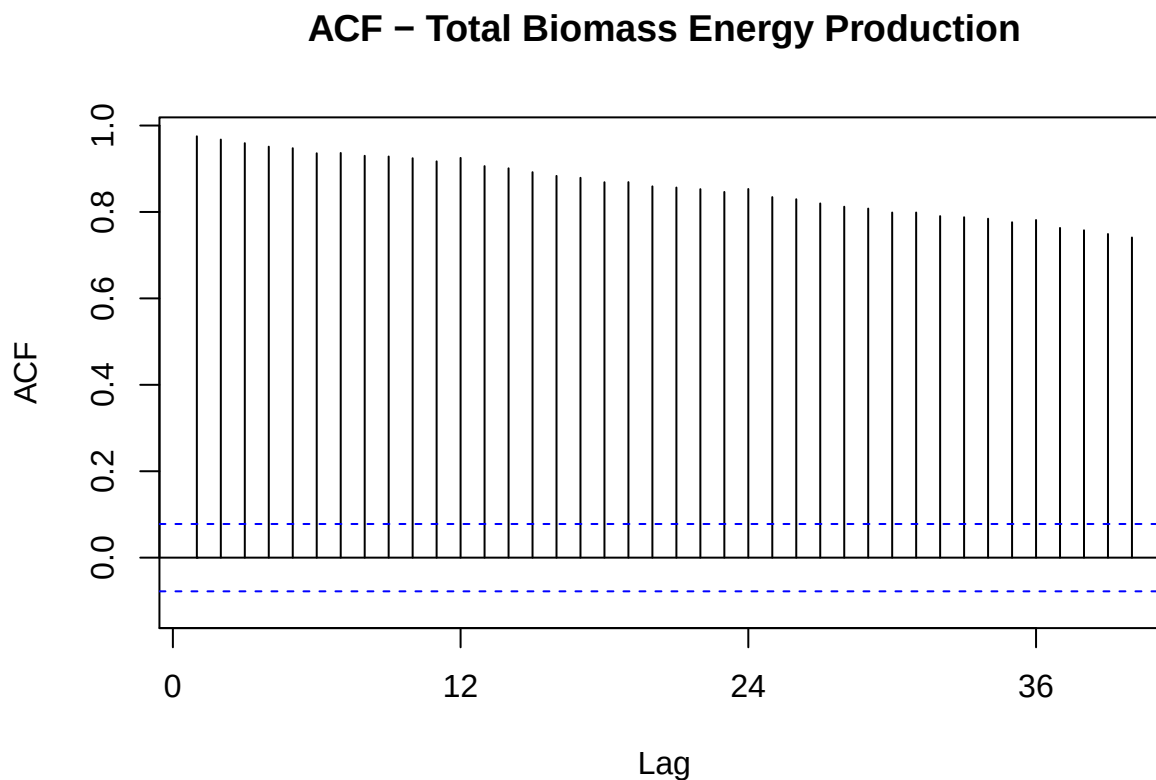
```
##                                Total Biomass Energy Production
## Total Biomass Energy Production      1.0000000
## Total Renewable Energy Production    0.9652985
## Hydroelectric Power Consumption      -0.1347374
##                                Total Renewable Energy Production
## Total Biomass Energy Production      0.96529851
## Total Renewable Energy Production    1.00000000
## Hydroelectric Power Consumption      -0.05842436
##                                Hydroelectric Power Consumption
## Total Biomass Energy Production      -0.13473742
## Total Renewable Energy Production    -0.05842436
## Hydroelectric Power Consumption      1.00000000
```

Looking at the correlation matrix, biomass and renewables are strongly correlated, while hydro shows almost no relationship with either. The biomass-renewables correlation is statistically significant given the sample size.

Question 6

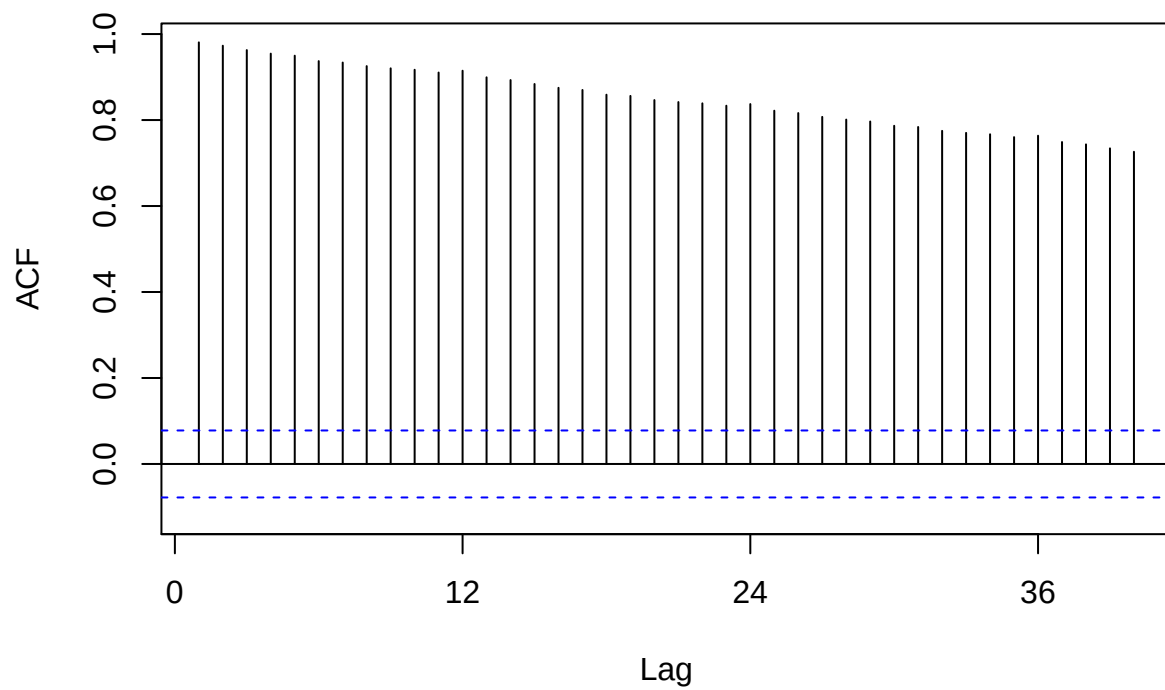
Compute the autocorrelation function from lag 1 up to lag 40 for these three variables. What can you say about these plots? Do the three of them have the same behavior?

```
#ACF up to lag 40 for the three series  
par(mfrow = c(1,1))  
Acf(energy_ts[, "Total Biomass Energy Production"], lag.max = 40,  
    main = "ACF - Total Biomass Energy Production")
```



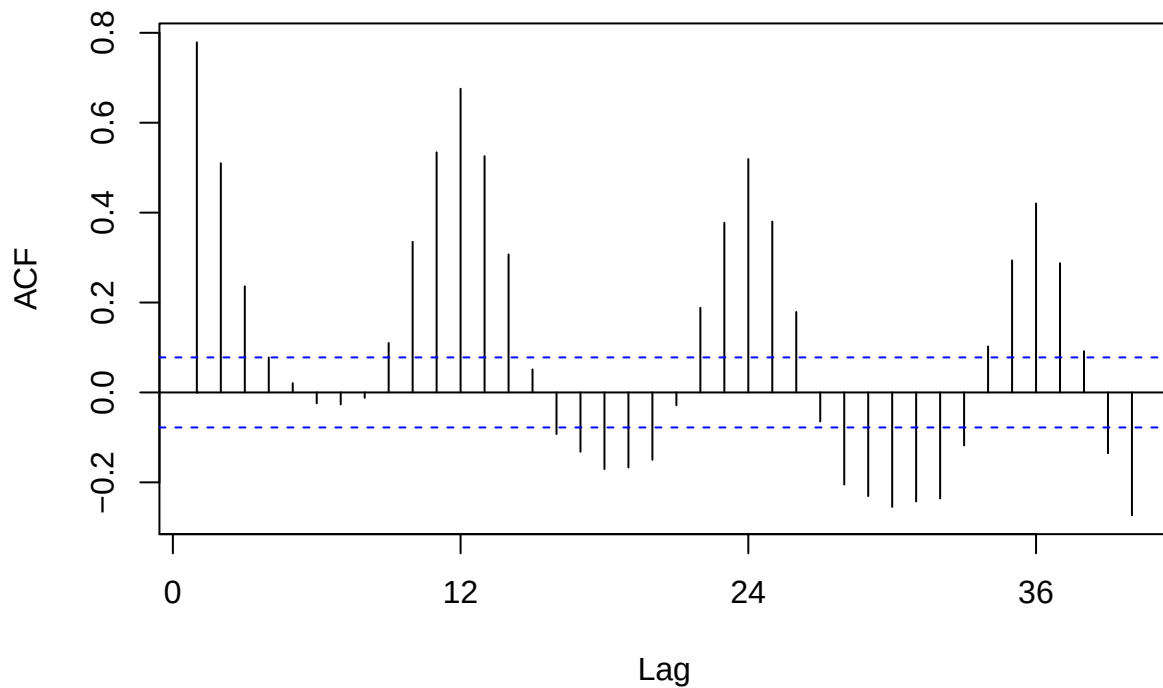
```
Acf(energy_ts[, "Total Renewable Energy Production"], lag.max = 40,  
    main = "ACF - Total Renewable Energy Production")
```


ACF – Total Renewable Energy Production



```
Acf(energy_ts[, "Hydroelectric Power Consumption"], lag.max = 40,  
     main = "ACF - Hydroelectric Power Consumption")
```

ACF – Hydroelectric Power Consumption



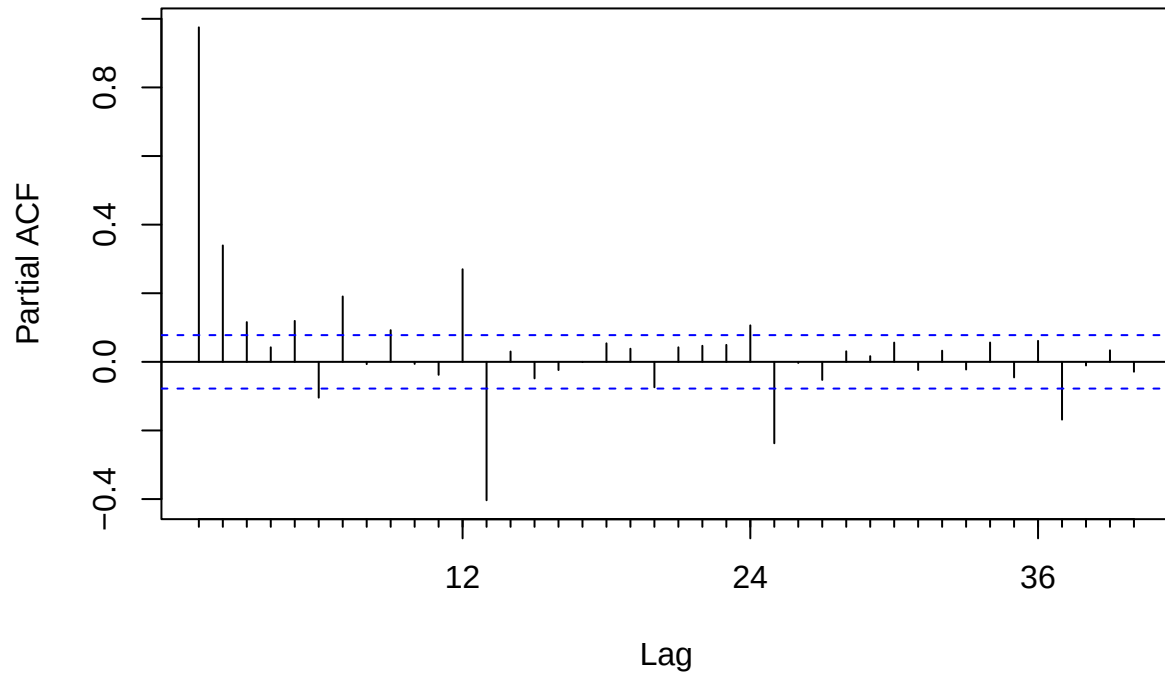
The ACFs for Biomass and Renewables shows on the plot both slowly and stay positive on all the lags, which is a sign of a strong trend and non-stationarity. Hydro is different, its ACF oscillates with clear peaks at lags 12, 24, and 36, which reflects the yearly seasonality in hydropower.

Question 7

Compute the partial autocorrelation function from lag 1 to lag 40 for these three variables. How these plots differ from the ones in Q6?

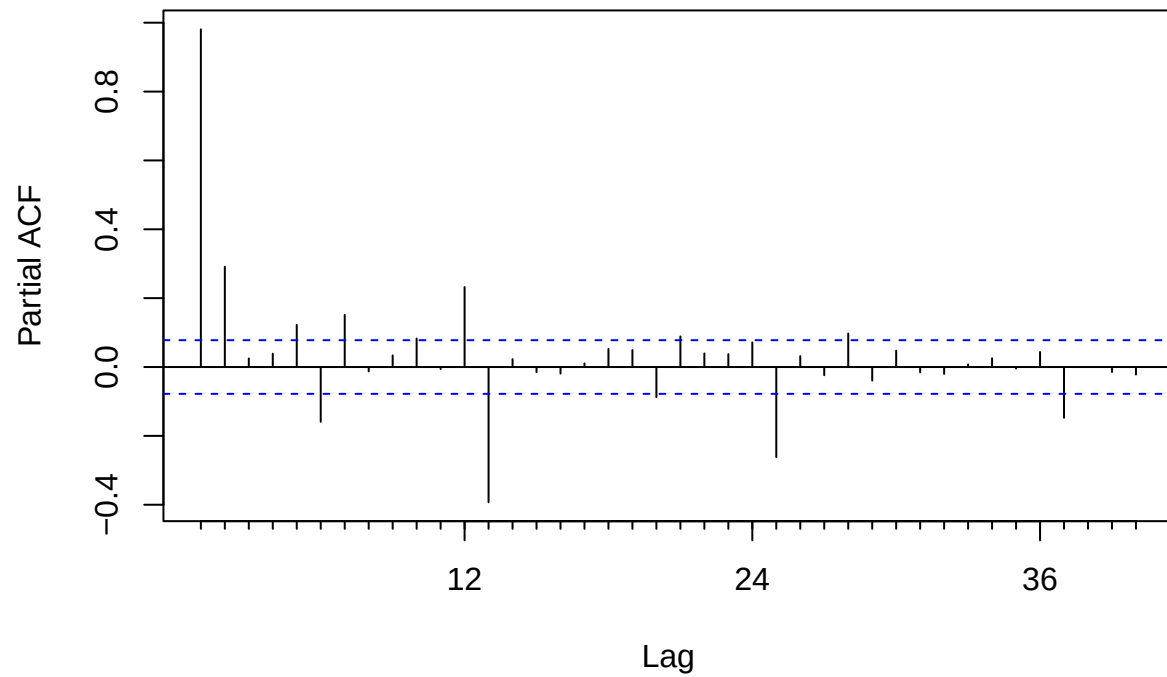
```
#PACF up to lag 40 for the three series
par(mfrow = c(1,1))
Pacf(energy_ts["Total Biomass Energy Production"], lag.max = 40,
     main = "PACF - Total Biomass Energy Production")
```

PACF – Total Biomass Energy Production



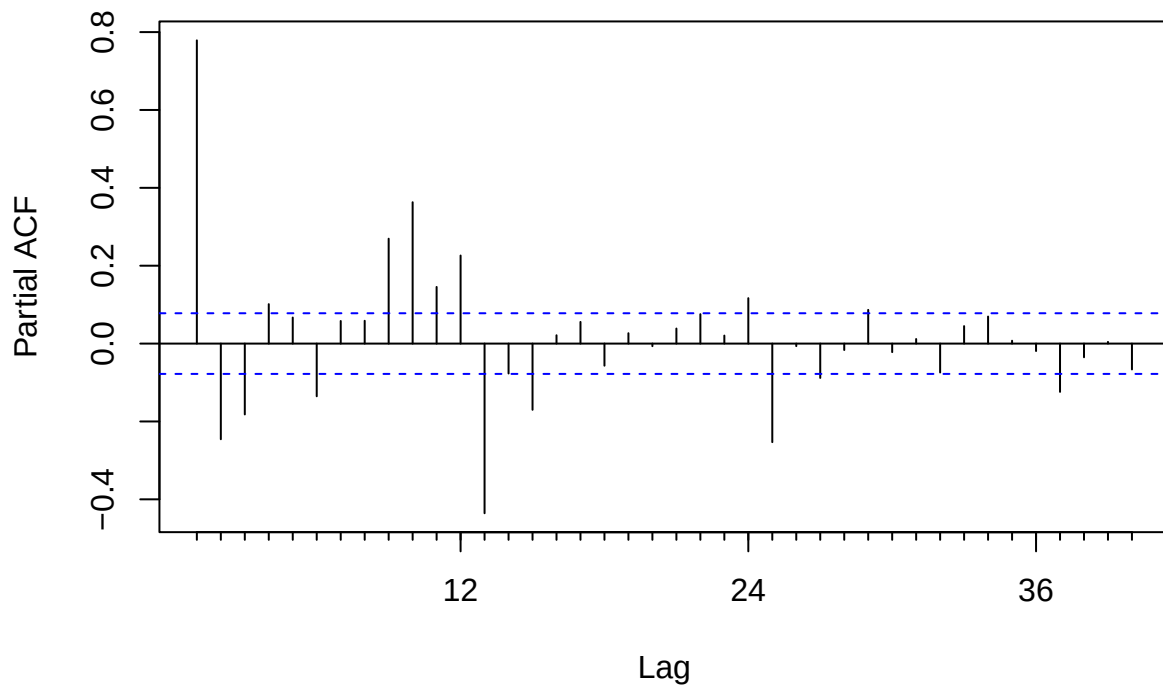
```
Pacf(energy_ts["Total Renewable Energy Production"], lag.max = 40,  
     main = "PACF - Total Renewable Energy Production")
```

PACF – Total Renewable Energy Production



```
Pacf(energy_ts["Hydroelectric Power Consumption"], lag.max = 40,  
     main = "PACF - Hydroelectric Power Consumption")
```

PACF – Hydroelectric Power Consumption



The PACFs show that most of the high correlations from Q6 collapse to just one or two significant lags. Lag 12 still stands out, especially for Hydro, which confirms the seasonal pattern. The main difference from Q6 is that the ACF reflects the total correlation including the indirect effect through earlier lags, while the PACF isolates the direct correlation at each lag.